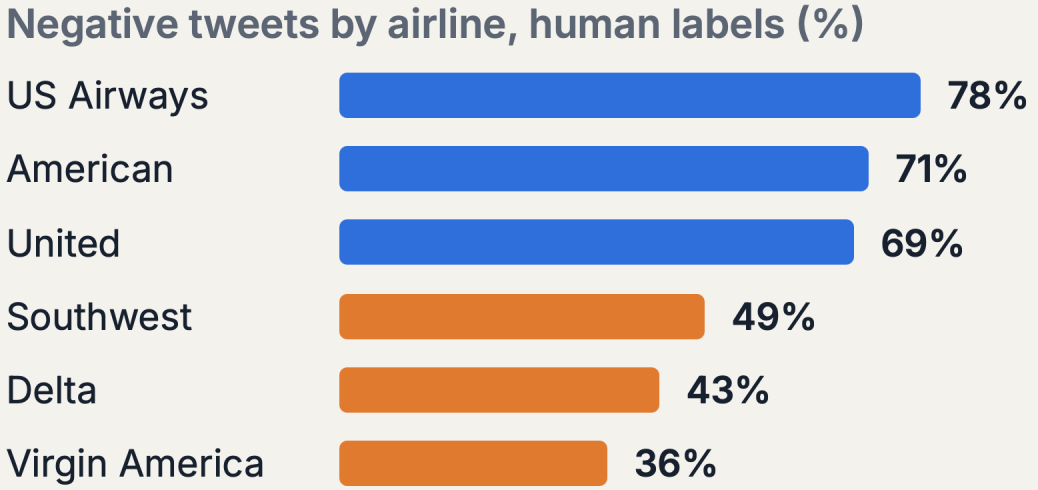
Exploring NLP Concepts for Social Media Analytics

Syntactic and semantic analysis of 14,640 airline tweets

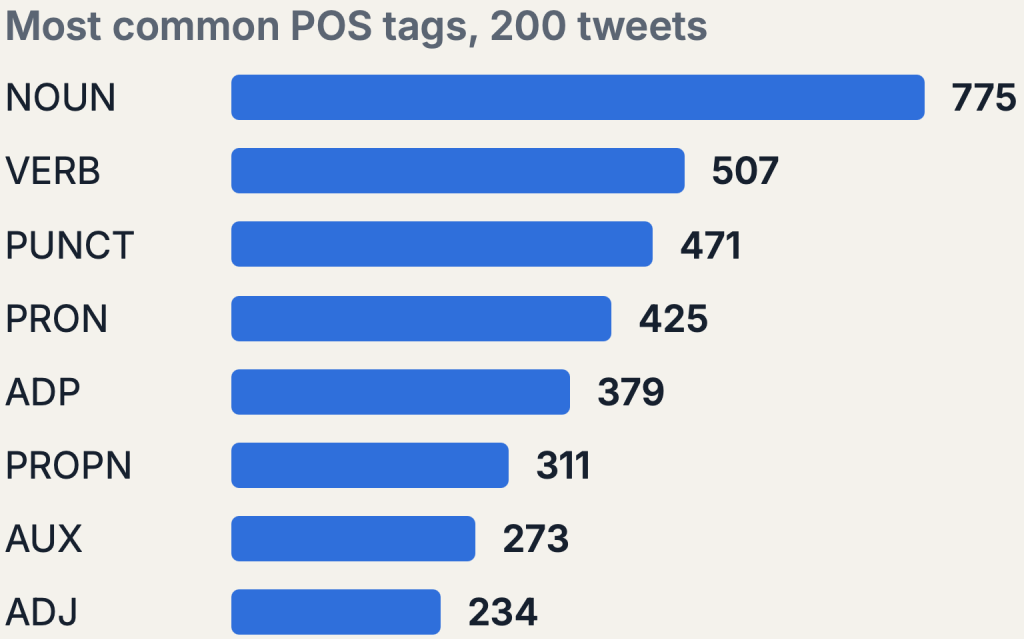
Name	Registration No.
Joy Melvine Okinyi	SCT213-C002-0099/2023
Maria Wangari Kanene	SCT213-C002-0029/2023
Lorna Ndanu	SCT213-C002-0127/2023
Vincent Kimutai	SCT213-C002-0053/2023
Stephen Koigi	SCT213-C002-0025/2023
Spencer Ngari	SCT213-C002-0117/2023

Topic 1: Preprocessing

- Dropped duplicates, URLs and @handles
- Kept negations such as not and no
- Lemmas stay readable; Porter stems give commerci
- Vocabulary: 10,972 tokens, 9,344 lemmas, 8,345 Porter stems



Topic 2: Syntactic analysis



Tweet #1

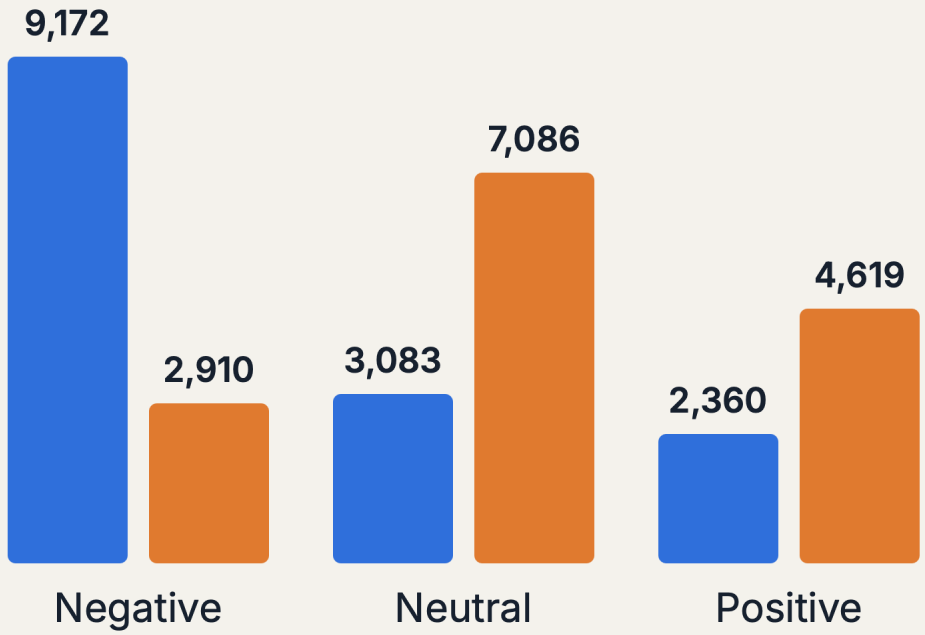
@united cleaning a regional jet takes an hour?

The tagger labelled @united as ADJ instead of a proper noun, so the parser attached cleaning to it as a modifier.

One tagging error distorts the whole parse.

Topic 3: Semantic analysis with TextBlob

Tweets per label: human (blue) vs TextBlob (orange)



43.3%

accuracy on 14,615 tweets

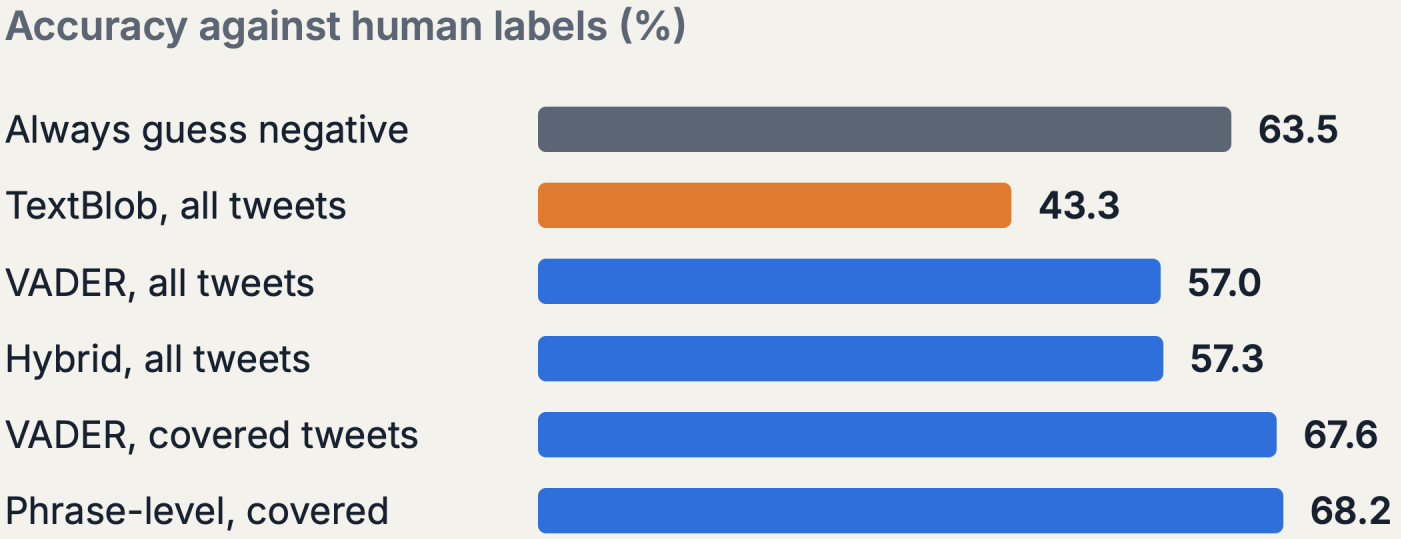
Caught only 29% of real negatives; labelled 7,086 tweets neutral

Topic 4: Sentiment by aspect

Mean phrase sentiment from dependency-based extraction, 14,294 tweets



Does syntax improve accuracy?



Covered = the 32.7% of tweets where a syntactic rule found an opinion. Gain over VADER is 0.6 points.

Conclusions

Syntax: where to look

Dependency rules link opinions to aspects, airlines and negations.

Semantics: how to feel

Lexicon scores give polarity but miss sarcasm and domain slang.

Limits

Rules cover 32.7% of tweets; accuracy gain is small.

Next: passive and implicit-complaint rules, and a domain-trained sentiment model.

Thank you!

Questions?

